

Carpathian OER Commons — Pathway to the Complete Guide

How the Complete Guide Gets Built, and How It Travels

Michel Garand | 2026-09-27 | Version v0.4 | CC BY-SA 4.0

Working paper of the Carpathian OER Commons, published as a working draft and open to co-creation. The project's child-protection rules apply in full: no child is named, dated, counted, or placed; the site is not named. Section 11 sets the rules that govern everything this pathway would publish.

Published 2026-09-27, beside the foundations paper. For publication, references to other writing about the place were made general, and the project's internal terms made plain; the content is unchanged.

This paper sits above the technical guides rather than beside them. It says what the complete guide consists of, how each part reaches the standard at which it can be published, and how it travels to other valleys — in Ukraine and across the borders the mountains do not recognise.

v0.4, 2026-09-27: the settlement group, the adult strand, and the pattern layer added from the open hamlet working paper, concept_open_hamlet_EN_v0_1.md. The map grows from thirty-four modules to forty.

1. What is being built

Two things are being built at once, and confusing them wastes both.

The first is a place: buildings, water, power, sanitation, a forest, a programme. It exists to serve the children who come and the household that hosts them, and it would be worth doing if no one ever copied it.

The second is a description of the place, accurate enough that people fifty or five hundred kilometres away can build their own version without repeating the expensive mistakes. That description is the guide.

The guide is not a by-product of the first thing. It is its own work, with its own standard, its own labour, and its own cost — and if it is treated as something to write up afterwards, it will not be written. What follows is the pathway to producing it deliberately.

The place is also becoming a settlement. The forest school serves adults too — retreat and continuing education — and around it a small hamlet is planned: households living there, with work, learning, and land held together. The hamlet is part of the place, and its description is part of the guide: the place's own pattern language, discovered and not imposed, and the modules that implement it. The reasoning is in the open hamlet working paper.

A model, not a template. Other initiatives in the Carpathians — Ukrainian, Romanian, Polish, Slovak, Hungarian — work in different regulations, currencies, soils, and histories. What travels is the method, the sizing, the order of decisions, and the record of what failed. What does not travel is the answer. A guide that pretends otherwise is an export, and this work is not for export; it is with.

2. What is taken from Open Source Ecology, and what is not

Open Source Ecology, founded by Marcin Jakubowski, set out to publish open blueprints for the fifty industrial machines it takes to run a small civilisation — the Global Village Construction Set — built and tested on its own farm, designed to be modular, repairable, and made locally at a fraction of commercial cost. Four of its habits are worth taking whole.

- **Documentation is the deliverable.** A thing built and not documented has not been finished. OSE treats a demonstrated, replicable production model as the completion metric, not a working prototype.
- **Modularity.** Independent parts that interoperate, so that a reader takes the two modules they need without swallowing the whole.
- **Lifetime design and serviceability.** Built to be repaired by the people who use it, with the documentation that makes repair possible.
- **Replication is the measure of success.** Not how good the original is, but how quickly others can build and improve on it.

Four things are not taken.

- **The machines.** This is not a fabrication project. The technologies here are a cistern, a reed bed, a wood stove, a battery, a nursery — mostly older than industry and mostly built by local trades.
- **The completeness claim.** No set of fifty anything. The guide covers what this site actually does, and says plainly where it stops.
- **The rhetoric.** No post-scarcity, no new economy, no civilisation-building language. The register here is a household in a war, describing what it built and what it cost.
- **Open by default, everywhere.** A hardware project can publish everything. This one cannot: children, a wartime location, and knowledge that is not ours to license all set limits. Section 11.

3. The unit: the module

Everything in the guide is a module with the same shape, so that a reader who has read one can navigate any other, and so that translation and revision stay tractable.

Every module carries:

- **Purpose** — what problem it solves, in two sentences, and one line of connections: what the module depends on and what it feeds.
- **Context of validity** — the conditions under which the design holds: climate, altitude, group size, season of use. The most important section, and the one usually missing from published projects.
- **Sizing method** — the arithmetic, with the input a reader must measure themselves, not our result.

- **Bill of materials** — quantities, with substitutions that are known to work.
- **Build** — sequence, trades needed, time, what happens if it is done in the wrong order.
- **Cost** — with currency, month, and place of the quotation. A cost without a date is a rumour.
- **Operation and maintenance** — daily, seasonal, annual; who is responsible.
- **Failure modes** — what breaks, what it looks like when it is going wrong, what it costs to fix.
- **What we got wrong** — the section that makes the guide worth reading.
- **Regulatory notes** — by country, as a table, each entry dated and attributed to whoever checked it.
- **Adaptation notes** — what a reader must change for a drier valley, a colder one, a smaller group.
- **Provenance and licence** — who wrote it, from whose knowledge, under what terms.
- **What to check** — every figure marked [set] or [assumed], and how it will be replaced.

Figures that more than one module sizes from — capacity, litres per person, the exchange rate — are declared once, in a site parameters file in standards/, and cited by name. A change there is carried through every module that cites it in the same commit.

The three technical guides already drafted — water, energy, sanitation — feed eleven rows of the component map between them, not three. The map's source column shows which. Converting them is the first task of the pathway, not a later tidy-up.

Patterns. Under its Connections line every module carries a pattern line: what it grows from and feeds among the place's own patterns, once their cards exist; which of Alexander's patterns it resonates with, as prompts and never as specification; and what it is in tension with. Three kinds of pattern are kept apart — the place's, named by the household, the co-creators, and the elders; Alexander's, used as prompts; and a module's own design rules, which stay inside the module.

4. Maturity: what may be claimed

Every module carries a level, stated at the top. Readers act on the level, not on the confidence of the prose.

Level	Meaning	May publish
L0	Idea, discussed	nothing
L1	Designed and costed, basis per line	design, marked untested
L2	Built here	build record, real cost
L3	Measured through a full season	performance numbers

Level	Meaning	May publish
L4	Built by someone else, elsewhere	their figures beside ours
L5	Improved by others, merged back	the improved version

Whether a cost is quoted, estimated, or assumed is a label on each cost line, not a level. A module at L1 may be costed entirely from estimates, and says so line by line.

The three existing guides sit at L1. Nothing in them should be read as tested, and the modules must say so on their face. The temptation the pathway has to resist is publishing L1 work in L3 language.

5. The component map

Source says where existing material lies: W, E, S for the water, energy, and sanitation guides with their section numbers; P for the programme proposal, published beside the foundations paper; 7a for the project's child-protection rules, held in its working knowledge; I for the infrastructure proposal, not yet located.

5.1 Ground and safety

Module	Status	Source
Shelter: standard, siting, certification	L0	E 9, W 2
Warning, communications, evacuation	L0	—
Fire protection and night lighting	L0	—
Medical room and quiet room	L0	—

5.2 Utilities

Module	Status	Source
Water: spring, cistern, treatment	L1	W 1–5, 8–9
Rainwater and nursery irrigation	L1	W 6, 8.5
Energy: PV, storage, generator	L1	E 1–4, 6–7, 9
Micro-hydro, if the stream allows	L0	E 5, 7.6

Module	Status	Source
Sanitation: terra preta toilets	L1	S 1–3, 8.1, 9–10
Greywater, reed bed, pond	L1	S 6–7, 8.5–8.6
Wood heat and hot water	L0	E 8, S 8.7

5.3 Buildings

Module	Status	Source
Sleeping accommodation, winter-capable	L0	—
Kitchen and dining to institutional standard	L0	—
Washroom block	L1	S 8.7–8.8
Covered gathering place and hearth	L0	—
Workshop and tool store	L0	—
Insulation and mountain building practice	L0	—

5.4 Land, forest, food

Module	Status	Source
Tree nursery: beds, shade, watering	L1	I
Biochar: kiln, quality, storage	L1	S 2, 8.2
Terra preta: fermentation to soil	L1	S 4–5, 8.3
Planting, succession, the mountain calendar	L0	P 4.2
Compost and site nutrient cycle	L1	S 5, 8.4
Kitchen garden and food storage	L0	—

5.5 Programme

Module	Status	Source
Pedagogy: the pathway and its stages	L1	P 3, 5

Module	Status	Source
Rhythm: the day, the session, the year	L1	P 4, 5
Pattern cards and the record	L1	P 3.1
Elders' strand and its consent practice	L1	P 4.4
Safeguarding: the two tiers and the gates	L2	7a, P 6
Mentor preparation	L0	—
Adult strand: retreat and continuing education	L0	—

5.6 Organisation

Module	Status	Source
Legal form and ownership	L0	—
Budgeting: capital and running	L1	W 8, E 7, S 8, I
Funder fit and application practice	L1	P, I
Cross-border partnership	L0	—
Documentation toolchain and standards	L2	standards/

5.7 Settlement

Module	Status	Source
Settlement form and siting, schematic	L0	—
Dwellings for resident households	L0	—
Paths, gateway, and access	L0	—
Common land and shared work	L0	—
Membership and self-governance	L0	—

Self-governance is designed threefold: those who teach decide pedagogy; all members decide the standing rules, each with one voice; the associations of those who produce and use decide the economy. Safeguarding sits in the rights sphere, beyond the reach of the other two.

Forty modules. Two at a level where the site's own experience backs them, sixteen at design stage, twenty-two untouched. Naming the empty rows is more useful than filling them with prose.

Across all seven groups runs a pattern layer — the place's own patterns, discovered, not imposed — carried in the pattern record and in each module's pattern line. It is a second way of reading the map, not a group in it.

Two L0 rows already have costed material in the guides — micro-hydro and wood heat — held at L0 because each waits on a measurement or a choice not yet made: the stream's head and flow, and the buildings the heat serves.

6. The pathway, in five phases

Phase 0 — Standards and repository. Fix the module template, the file naming, the versioning, the units and currency conventions, the image rules, and where the repository lives. Convert the three existing guides to module shape. Nothing is published yet. Weeks, not months, and it is the phase most often skipped.

Phase 1 — Document what exists. Before any new building: measure and write up what the site already has. The spring's yield. The existing flush toilets and where they discharge. The buildings, their construction, their heat loss. This is the baseline every later figure is read against, and it can only be captured once.

Phase 2 — Build and measure. Each module moves L1 to L2 as it is built, with the build recorded as it happens — dated photographs, real invoices, the day the mason said the wall needed to be thicker. Then a full season of measurement before any performance claim: water in the dry month, battery through December, the reed bed under snow.

Phase 3 — Publish. English settles first; the Ukrainian sibling follows and is corrected over time rather than held back. Publication is module by module as each reaches L2, not a single release at the end. A module that is honest about being L1 can be published early and marked.

Phase 4 — Invite replication. One or two sites first, not a launch. A site that is already building something and needs two of our modules is worth more than fifty readers. Their figures come back into the module beside ours, which is what L4 means.

Phase 5 — Commons governance. Once other people's work is in the guide, it needs a way of deciding what gets merged, who holds the repository, and what happens if the original site stops. Deferred deliberately — do not build governance for a commons that has one member.

7. Toolchain

The guide is written in the same markdown standard as everything else in this project, checked by the same script, and stored in a version-controlled repository — git, hosted somewhere the household can reach and does not pay for. Other writing about the place stays where it is; the guide is a separate thing with a separate rhythm, because testimony and technical documentation age differently.

- **Text:** markdown, ERDPULS standard, one file per module, checker before commit.
 - **Drawings:** SVG or FreeCAD, source files committed beside the exports. A PDF of a drawing is not a drawing.
 - **Measurements:** plain CSV, one series per file, with a header stating instrument, place at valley resolution, and who took it.
 - **Photographs:** metadata stripped, filenames neutral, reviewed against Section 11 before they go anywhere near a server.
 - **Translation:** English canonical; other languages as siblings with a stated revision date, never blocking publication.
 - **Releases:** a dated snapshot when a group of modules reaches L2, so that a replicating site can cite a fixed version.
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8. Licensing, and what cannot be licensed

Documents: CC BY-SA 4.0, as everything else in this project. Share, adapt, attribute, keep it open.

Designs and drawings: a hardware-appropriate licence rather than a document licence — CERN-OHL-S is the usual choice for open hardware that must stay open. The distinction matters the first time someone builds from a drawing and improves it.

Measurement data: released openly, ideally with no conditions beyond attribution, since data that comes with restrictions stops being compared.

Attribution is joint. The philosophy document already carries two hands. The technical modules will carry more: the household, the mason, the electrician, the elders. An open licence that quietly attributes shared work to one author is a theft dressed as generosity.

Not ours to license. Hutsul practice — forms, songs, craft, the knowledge of a particular valley — is not automatically open because we wrote it down. Anything of that kind enters the guide only with the consent of the people whose knowledge it is, on the terms they set, and some of it will not enter at all. The right default is to describe our own use of a practice and point to its holders, rather than publish the practice as if it were ours to give.

9. Across the borders

The Carpathians run through Ukraine, Romania, Poland, Slovakia, Hungary, Serbia, and Czechia, and a cross-border frame for the range already exists in the Carpathian Convention. That frame is worth understanding before inventing a network: it has working groups, funded programmes, and a habit of cooperation that predates this project. Check its current standing before citing it in any application.

What travels well across those borders: sizing methods, the order of decisions, failure modes, the maturity discipline, the safeguarding architecture.

What does not travel and must be localised by whoever is there: regulation — sanitary, building, water, child protection, all national; prices and trades; species and planting calendars; the language of the programme; the relationship with a local authority, which is always personal.

Practical consequence for the module template: the regulatory note is a table with one row per country, each row dated and signed by whoever checked it, and empty rows stay visibly empty. A Romanian reader must be able to see at a glance that nobody has checked Romania.

Translation policy: English canonical, Ukrainian close behind. Polish, Romanian, Slovak, and Hungarian only when a partner in that country takes on the translation as their own work — a machine-translated Romanian module with no Romanian reader is worse than none, because it looks checked.

10. How another site would use this

The replication protocol, written from the reader's side:

1. **Self-assessment.** Group size, season of use, altitude, water source, grid, legal status. Half a page, which decides which modules apply.
2. **Measure first.** Each module names what must be measured before it can be used. A site that skips the spring measurement and builds our cistern has not replicated anything.
3. **Take modules, not the model.** A site that needs only sanitation takes sanitation. **Walk your own settlement** with Alexander's patterns as prompts, and begin your own pattern record, as in Section 4 of the open hamlet paper. Your knowledge-holders name your patterns; ours are not yours.
4. **Localise the regulation.** Their row in the table, checked by them, sent back.
5. **Build, record, return figures.** Real costs in their currency and month; what they changed and why.
6. **Twinning, if wanted.** A working relationship between two households beats any amount of documentation, and the documentation exists mostly to make the first conversation shorter.

A failure log is part of the offer: sites that tried something from the guide and abandoned it, with the reason. The absence of such a log in most published projects is why most published projects cannot be trusted.

11. Safeguarding governs everything here

The guide is a published artefact, so the project's child-protection rules apply to every line of it, and the aggregation test applies across modules: a drawing in one, a photograph in another, and a date elsewhere can together place a site that no single document placed.

- **No child in any module.** Not in a photograph, not in a caption, not in an example, not in a headcount, not as texture. Programme modules describe standing format only.
- **Publish the method, not the place.** Site plans in a published module are schematic — relationships only, without distances or the terrain that identifies a valley, until the decision on naming the site. The site is the hamlet: dispersed homesteads on a slope, drawn with distances, place a valley. Construction drawings of a single structure keep their dimensions.
- **Photographs** show structures and systems, no faces, no signage, no distinctive ridgelines, no metadata.
- **Regulatory and application material** that names an entity travels to funders, not to the repository.
- **Pattern cards** stay in the workshop. Only the yearly narrative of the place's pattern language is published, at valley resolution.
- **Wartime review before every release**, not once at the start. What was safe to publish in one season may not be in the next.
- **Gate 1 throughout.** The household decides what of its own life is described, module by module.
- **A line held beyond the rules.** Children, wartime, a named doorstep, a fixed week, and family circumstance, in one piece, are never published, whatever the rules at the time allow.

The guide is publishable because infrastructure is publishable. The programme modules are the difficult ones, and the right instinct there is to publish the architecture of the safeguarding and very little of the programme's daily life.

For a replicating site, which will not see the project's working rules: Tier 1 — no child named, dated, counted, or placed, ever; Tier 2 — anything more only through consent gates; Gate 1 — each household decides what of its own life is described.

12. Who pays for the documentation

Documentation time is real work — roughly a day per module to draft, and more to check, draw, and translate. It is invisible in most budgets, which is why most projects do not have documentation.

Three routes, not exclusive:

- **Inside the infrastructure budget.** A documentation line of five to eight percent of capital, justified as open educational resources, which several funders fund willingly.
- **As its own small grant.** Open knowledge, digital commons, and educational resource programmes fund exactly this, and the deliverable is unusually easy to verify.
- **In kind.** The household's time, counted as contribution at the same rate as toloka labour, so its value appears rather than vanishing.

The guide is not sold and is not a product. If someone wants more than the guide gives — a person on site, a design for their valley — that is a separate arrangement and does not close the guide.

13. The metric of completion

Borrowed from OSE and narrowed: **the guide is complete when another site has built from it, in another valley, and their figures are in it beside ours.**

Not when every module is written. Not when it is translated into six languages. Not when it is published. A guide with six modules and one honest replication is further along than thirty modules read by nobody.

14. The first twelve months

A sequence, not a schedule. Each step assumes the one before.

1. Fix the module template, the repository, and the site parameters. Convert the three guides into the eleven rows they feed.
2. Baseline the site: spring yield in the dry month, existing sanitation and its outfall, buildings and their condition, the stream if there is one. Walk the settlement with Alexander's patterns as prompts, and begin the pattern record.
3. Send the written enquiries: establishment category, the spring's protection zone, discharge to ground, a fire-fighting reserve, a water-use permit for the stream. File each answer against the regulatory row of the module it governs.
4. Write the safeguarding module first among the programme modules — it is at L2 already and it governs the rest.
5. Publish the utility modules at L1, marked as untested, in English and Ukrainian.
6. Build the safety and utility layers; record as built, not after.
7. Measure one full season, including one winter.
8. Move the built modules to L2, then L3 where a season supports it.

9. Find one other site. Not a network, not a launch. One.

15. What has to be decided

- Whether the guide names the site at all, ever — and if so, at what point in the war.
 - Decided 2026-09-27: the repository. A private workshop on GitHub under ubecccommon, publication as a collection in ubecccommon.github.io; every clone is a full local copy for when the connection fails.
 - Which hardware licence, decided before the first drawing rather than after.
 - Who the co-authors are, named on each module, and how the elders' material is handled.
 - Whether documentation is funded inside the infrastructure budget or separately.
 - Which country a partner might come from first, since that decides the second language after Ukrainian.
 - Decided 2026-09-27: the programme proposal and the philosophy are published as working drafts, in standing format, beside the foundations paper.
 - Legal form and ownership, before anyone lives on the site. One option to weigh, not decided: land held in trust, out of the market.
 - Each resident household's own Gate 1, before its life is described.
 - How safeguarding supervision extends to retreat guests and residents.
 - How much of the adult strand is published.
 - Decided 2026-09-27: released modules carry the author's name and the initiative's name.
 - Decided 2026-09-27: capacity, session length, and use days may be published, as standing format and never beside a date.
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Release record

- **Gate 1:** 2026-09-27, recorded by Michel Garand — published beside the foundations paper as a working draft, open to co-creation and to change.
 - **Wartime review:** 2026-09-27, Michel Garand — names no place; references to other writing about the place made general; accepted.
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